Laboratorio No.7 – Ejercicio 1.1

Esteban de la Roca 24171
Martin Villatoro 24033

1)

a)

$C \Rightarrow \varepsilon$

$A \Rightarrow C \Rightarrow \varepsilon$

$B \Rightarrow A \Rightarrow \varepsilon$

$N_\varepsilon = \{S, A, B, C\}$

$S_0 \to S \mid \varepsilon$

$S \to 0A0 \mid 00$

$S \to 1B1 \mid 11$

$S \to BB \mid B$

$S_0 \to S \mid \varepsilon$

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$

$A \to C$

$B \to S \mid A$

$C \to S$

b)

$S_0 \to S$

$S \to B$

$A \to C$

$B \to S$

$B \to A$

$C \to S$

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$S \Rightarrow B \Rightarrow A \Rightarrow C \Rightarrow S$

$S \equiv A \equiv B \equiv C$

$S_0 \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \varepsilon$

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$A \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$B \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$C \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

c)

S_0, A, B, S, C

$P = \{S_0, S, A, B, C\}$

$S_0 \Rightarrow 0A0$

$S_0 \Rightarrow 1B1$

$S_0 \Rightarrow BB$

$Alc = \{S_0, A, B\}$

$S, C \notin Alc$

S, C

$S_0 \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \varepsilon$

$A = 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$B \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

d)

$X_0 \to 0$

$X_1 \to 1$

$0A0$

$Y = A X_0$

$0A0 = X_0 Y$

$1B1$

$Z \to BX_1$

$1B1 \Rightarrow X_1 Z$

$S_0 \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \varepsilon$

$A = 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$B \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

2)

$S \to aAa \mid bBb \mid \varepsilon$

$A \to C \mid a$

$B \to C \mid b$

$C \to CDE \mid \varepsilon$

$D \to A \mid B \mid ab$

a)

$S \Rightarrow \varepsilon$

$C \Rightarrow \varepsilon$

$A \Rightarrow C \Rightarrow \varepsilon$

$B \Rightarrow C \Rightarrow \varepsilon$

$D \Rightarrow A \Rightarrow \varepsilon$

$N_\varepsilon = \{S, A, B, C, D\}$

$S_0 \to S \mid \varepsilon$

$aAa \Rightarrow aAa \mid aa$

$bBb \Rightarrow bBb \mid bb$

$CDE \Rightarrow CDE \mid DE \mid CE \mid E$

$S_0 \to S \mid \varepsilon$

$S \to aAa \mid aa \mid bBb \mid bb$

$A \to C \mid a$

$B \to C \mid b$

$C \to CDE \mid DE \mid CE \mid E$

$D \to A \mid B \mid ab$

b)

$S_0 \to aAa \mid aa \mid bBb \mid bb \mid \varepsilon$

$S \to aAa \mid aa \mid bBb \mid bb$

$A \to a \mid CDE \mid DE \mid CE$

$B \to b \mid CDE \mid DE \mid CE$

$C \to CDE \mid DE \mid CE$

$D \to a \mid b \mid ab \mid CDE \mid DE \mid CE$

c)

$S_0 \to aAa \mid aa \mid bBb \mid bb \mid \varepsilon$

$A \to a$

$B \to b$

d)

$A \to a, \quad B \to b$

$aAa \Rightarrow AAA, \quad bBb \Rightarrow BBB$

$X_A \to AA, \quad X_b \to B$

$S_0 \to AX_A \mid AA \mid BX_B \mid BB \mid \varepsilon$

$X_A \to AA$

$X_B \to BB$

$A \to a$

$B \to b$